

Project Brief: Working Document

Digital Audio Player • Rev 11

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1. Summary

A portable digital audio player designed from scratch by Alexander Loker: schematic, PCB, simulation, firmware and enclosure. The device is intended as a genuine end product (small, pleasant to use, and measurably good), and the project closes two deliberate skill gaps: custom MCU board design, and analog and mixed-signal design backed by simulation.

#	Objective	How it is satisfied
1	Build an audio player worth using	Portable microSD player, FLAC and MP3, single-ended 3.5 mm output tuned for IEMs, 2.4 in display.
2	Close the MCU-board knowledge gap	Custom ESP32-S3 board from the bare chip: power tree, boot straps, SDMMC, USB, I2S, I2C charger control.

#	Objective	How it is satisfied
3	Work with data converters	ES9038Q2M current-output DAC with a discrete I/V and filtering chain.
4	Work with analog signals	I/V conversion, differential summing, reconstruction filtering, output impedance control, low-noise power.
5	Simulate where simulation earns its place	LTspice for the buck-boost converters, inverter and LDOs; signal-chain simulation where it informs a decision.
6	Learn	Each design decision documented and verified by measurement on the finished board.

Changes since Rev 8: schematic complete; MCU without PSRAM, 1-bit SDMMC, external flash; dedicated DAC analog rail; charge-pump backlight driver; right-angle EN12 encoder replaces the front buttons; display panel selected; CE and INT are the only charger GPIOs.

2. Locked Scope

Decision	Choice	Rationale
MCU	ESP32-S3 base chip, QFN-56; no PSRAM; 8 MB quad SPI flash (GD25B64EWIGR)	USB-OTG device, SDMMC, I2S, 240 MHz. ESP-IDF / FreeRTOS.
DAC	ES9038Q2M	Differential current output; I2S master with the DPLL off.
Clocking	Two tri-state oscillators on XI: 22.5792 MHz (ECS) and 24.576 MHz (Abracon)	Maximum sample rate 176.4 / 192 kHz.
Source	microSD, 1-bit SDMMC, card detect; local files only	No Bluetooth, no streaming, no USB Audio Class.
Formats	FLAC 16/44.1 minimum, MP3	Decoded on the MCU to PCM.
USB-C role	Charging and file transfer only	TinyUSB MSC exposes the microSD as a drive.
Output	3.5 mm TRS, single-ended, DC coupled (SJ1-3535NS)	Sub-1 Ω output; firmware mutes on a rail fault.
Load target	Simgot EW300 (28 $\Omega \pm 15\%$); 16 to 32 Ω IEMs	Unforgiving of noise floor and output impedance.
Maximum output	0.6 V RMS (0.85 V peak)	114.6 dB SPL peaks on the EW300.
I/V and summing op-amp	TI OPA1612AIDRGR $\times 3$	Four I/V channels and two summing channels.
Output amp	TI OPA1622IDRCR	Dual unity-gain follower; thermal pad to V-; EN for shutdown.
Analog rails	± 3.3 V; DAC AVCC / VCCA on a dedicated +3.3 V LDO (+3V3_DAC)	The DAC output scales with AVCC; no op-amp current on its supply.
Display	BuyDisplay ER-TFT024IPS-3: 2.4 in IPS, 240 $\times$ 320, ST7789V, SPI, no touch	Glass fits inside the battery footprint.
Backlight	ROHM BD1604MUV charge pump from SYS, two levels via switched ISET	No inductor; DC LED current (no audio-band PWM).
Controls	TT Electronics EN12 right-angle encoder with push switch (rotate: up / down, press: select); three TL1014 side switches	Encoder and buttons are better in-pocket; frees UART0 for debug.
Battery	Jauch LP605060JU+PCM, 1S1P 3.7 V, 1850 mAh	63.0 $\times$ 51.0 $\times$ 6.6 mm envelope, integrated PCM.

Decision	Choice	Rationale
Charger	TI BQ24250RGET, I2C, CE and INT to GPIOs; EN1 / EN2 strapped (1 A input limit)	Charge at 500 mA (0.27C); TS gives hardware JEITA.
Analog pre-regulator	LTC3440, SYS → 3.9 V, 1.2 MHz	Feeds two LT3042 and the LT3462A inverter.
Negative pre-regulator	LT3462AES6, +3V9 → -3.9 V, 2.7 MHz	Dual-inductor inverter.
Analog LDOs	LT3042 ×2 (+3V3_ANALOG, +3V3_DAC) / LT3093 (-3V3), ±3.32 V	Ultralow noise; current limits 125 / 126.6 mA.
Digital rail	LTC3440, SYS → 3.318 V, 600 kHz, always on	Separate branch from SYS.
Fuel gauge	Voltage-based	Load-compensated lookup table.
EDA / CAD	Altium (AL-PN library), SolidWorks	Existing toolchain.

Source: [17]

3. Out of Scope

Item	Status	Note
4.4 mm balanced output	Dropped	Would require four amplifier channels.
Bluetooth / A2DP	Dropped	ESP32-S3 is BLE-only.
USB Audio Class	Deferred	Full-speed USB caps UAC1 around 24-bit / 96 kHz.
Coulomb counting	Rejected	Voltage-based gauging adopted.
BC1.2 / USB-C PD, CC sensing	Rejected	1 A input limit by strap; 5.1 kΩ CC pull-downs only; VDPM protects weak sources.
PSRAM, 4-bit SDMMC	Dropped	Pins used elsewhere; 1-bit SD outpaces Full-Speed USB and playback needs.
microSD power gating	Dropped	No load switch; card idle current counted in standby.
Touchscreen	Rejected	Encoder and buttons.
Line-in / recording	Deferred	Nothing forecloses it.
DSP / parametric EQ	Deferred	Added in firmware later.
Vented enclosure / cooling	Rejected	Dust, moisture, acoustic noise.

4. System Architecture

Signal chain: microSD → SDMMC (1-bit) → ESP32-S3 (decode to PCM in internal SRAM) → I2S → ES9038Q2M → I/V → differential summing and MFB low-pass → OPA1622 follower → 3.5 mm TRS. A local oscillator drives XI; the DAC generates BCLK and LRCLK.

Power chain: Li-Po → BQ24250 (SYS) → LTC3440 3.9 V → LT3042 (+3V3_ANALOG: op-amps, oscillators), LT3042 (+3V3_DAC: DAC AVCC / VCCA), LT3462A → LT3093 (-3V3). Digital LTC3440 3.3 V from SYS. Backlight charge pump from SYS. Single ground plane, analog and digital separated by placement.

4.1 Digital Subsystem

ESP32-S3 without PSRAM; decode on one core, UI and I2S on the other. 8 MB quad SPI flash on VDD_SPI (3.3 V by the GPIO45 strap). microSD in 1-bit SDMMC with card detect. UI: SPI TFT (display off during playback), EN12 right-angle encoder with push switch, side buttons; UART0 on test pads for a debug console. Charger over I2C with CE and INT on GPIOs; EN1 / EN2 strapped. Every GPIO is assigned (MCU document, Section 3.3).

4.2 Clocking

DAC as I2S master, DPLL off, MCLK = $128 \times f_S$. Two oscillators joined at XI through 22 Ω ; 10 k Ω pull-downs hold both off; firmware enables one after the analog rails are up.

4.3 Analog Chain

I/V: four OPA1612 channels, R_f 499 Ω , C_f 1.5 nF (213 kHz). Summing: differential MFB Butterworth, gain 0.441, f_0 69.4 kHz. Chain loss at 20 kHz $\approx$ 0.068 dB; $\approx$ 60 dB at 1 MHz. COG throughout the signal path.

4.4 Output Stage

OPA1622 follower with output-inclusive compensation ($4.7 \Omega / 499 \Omega / 1.5 \text{ nF}$): $Z_{\text{out}} \approx 0.44 \Omega$ at 20 kHz. Enable divider 1.57 V, 2.6 ms ramp. DC-fault protection by firmware on LDO power-good. ESD122 on tip and ring; plug detect on the tip switch.

4.5 Power Tree

LDO dropout sets the 3.9 V rail; the 3.9 V LTC3440 is switched off in standby (OPA1612 has no shutdown). The DAC reference supply has its own LT3042 so op-amp signal current never flows through it. Switchers: charger 3 MHz (plugged in only), LTC3440 1.2 MHz and 600 kHz, LT3462A 2.7 MHz, backlight charge pump $\approx$ 1 MHz only below $\approx$ 3.5 V battery.

4.6 Display Integration

ER-TFT024IPS-3 on a 0.5 mm FPC (F32D). Backlight: BD1604MUV from SYS, four sinks on LEDK1 to LEDK4, 4.8 or 19.6 mA per LED set by switching ISET with a 2N7002; EN on GPIO48, level on GPIO6.

Source: [15], [16], [17]

5. Target Specifications

Parameter	Target	Comment
Output connector	3.5 mm TRS, single-ended	DC coupled
Rated load	EW300, 28 $\Omega \pm 15 \%$	16 to 32 Ω supported
Output impedance	< 1 Ω , 20 Hz to 20 kHz	0.44 Ω at 20 kHz
Maximum output	0.6 V RMS	114.6 dB SPL peak on the EW300
Output noise	< 3 μV RMS, A-weighted	Hiss floor
THD+N	< -105 dB at 1 kHz, 32 Ω	Dedicated DAC reference rail supports this
Crosstalk	< -80 dB at 1 kHz	Shared ground return is the limit
Passband	Within 0.1 dB to 20 kHz	0.068 dB calculated
Formats	FLAC 16/44.1 and 16/48, MP3	Up to 192 kHz stretch goal
Battery life	$\geq$ 7 h playback	Power Budget Rev 7: $\approx$ 7.8 h derated
Charge time	$\approx$ 4.5 to 5 h at 500 mA	Including CV taper

Parameter	Target	Comment
Enclosure	$\approx 56 \times 75 \times 13.2$ mm	Driven by the cell envelope

6. Simulation Plan

Power tree: LDO PSRR, noise and output impedance complete for the LT3093; LT3042 noise sweep complete; LT3462A transient complete; 3.3 V LTC3440 compensation verified (author). Remaining: start-up and UVLO sequencing, 3.9 V LTC3440 at the 263 mA load. Signal chain: summing response and output stability complete; remaining: output impedance vs frequency, I/V noise and stability against DAC output capacitance, full-path noise, gain structure across the volume range.

7. Risk Register

Risk	Impact	Mitigation
Battery 28-week lead time	Stall if stock lapses	Purchase two units.
Runtime margin	6.5 h with 20 % margin	Measure the display; reserves: MCU clock, analog rails.
Charge time near the safety timer	Charging aborts early	Set the 9 h timer via I2C.
Charger switching noise	Noise floor while plugged in	Placement, shielded inductor, minimal loop area.
LT3462A switch current	Negative rail clips	Simulated 0.282 A vs 300 mA minimum limit.
3.9 V LTC3440 load	≈ 263 mA vs 250 mA design	Re-check load step or accept.
DC-coupled output fault	Destroys an IEM	Firmware monitors LDO power-good and drives EN low.
SD read latency	Audible dropouts	Ring buffer in SRAM sized against worst-case latency.
USB VBUS sensing	Pull-up enabled with no host	Firmware uses charger INT / status (MCU FW13).
USB MSC and playback contending	Filesystem corruption	Enumeration stops playback and releases the volume.
Standby drain	≈ 18 mW, ≈ 16 days	Accepted; SD not gated, digital rail always on.

8. Open Questions

#	Question
1	Volume control method: digital attenuation in the DAC, or analog ahead of the output stage. Affects the noise floor at low levels.
2	Measurement equipment: a -105 dB target needs an analyser floor below it.

9. Revision Log

Rev	Date	Change
6 to 8	2026-09-24 to 27	Clocking, I/V, grounding; summing, output stage, power figures.
9	2026-10-03	Schematic complete. Scope updated: no PSRAM, 1-bit SDMMC with card detect, 8 MB flash, dedicated DAC rail (+3V3_DAC), BD1604MUV backlight, EC12D encoder, ER-TFT024IPS-3 display, charger CE / INT GPIOs with strapped EN1 / EN2. Risks updated (load, VBUS, standby). Closed: display vendor.
10	2026-10-03	DAC analog rail named +3V3_DAC; oscillators confirmed on +3V3_ANALOG.
11	2026-10-04	Front UP / DOWN switches removed; encoder changed to TT Electronics EN12-VS11AF18 (right angle). Battery connector JST GH (BM03B-GHS-TBT). UART0 freed for a debug console.

10. Sources

1. ES9038Q2M Datasheet v1.4 (ESS Technology)
2. ECS-2520MVLC Datasheet (ECS Inc.)
3. ASTX-H11 Datasheet (Abracon)
4. OPA1612 Product Page and Datasheet (Texas Instruments)
5. OPA1622 Datasheet, SBOS727B (Texas Instruments)
6. Analog Engineer's Circuit Cookbook: Amplifiers, MFB Low-Pass Filter (Texas Instruments)
7. SJ1-353XNS Series Datasheet (Same Sky)
8. ESD122 Product Page and Datasheet (Texas Instruments)
9. bq2425x Product Page and Datasheet (Texas Instruments)
10. TPD2E2U06-Q1 Product Page and Datasheet (Texas Instruments)
11. LTC3440 Product Page and Datasheet (Analog Devices)
12. LT3462/LT3462A Datasheet (Analog Devices)
13. LT3042 Datasheet (Analog Devices)
14. LT3093 Datasheet (Analog Devices)
15. [BD1604MUV / BD1604MVV Datasheet \(ROHM\)](#)
16. [ESP32-S3 Hardware Design Guidelines \(Espressif\)](#)
17. MCU, Peripherals and subsystem working documents (current revisions)